

System Requirements

- Operating System: macOS
- CPU: ARM64 (Apple Silicon)
- Compiler: Apple clang 17.0.0 (clang-1700.4.4.1)
- Libraries: GNU Scientific Library (GSL)

Tested Environment

- macOS 15.6.1 (Sequoia)
- GSL 2.7.1

Run model calibration:

```
./exec -s 1 -verbose 0 -rerun 0 -e .
```

Table 1: Command line arguments

Argument	Type	Description
-s	int	Random seed
-verbose	int	Parameter for diagnostic checks • 1: print diagnostic checks • 0: suppress them
-rerun	int	Controls whether to perform model calibration or run simulations using known parameters. Specifically: • 0: perform model calibration using the Metropolis–Hastings MCMC procedure. • !=0: load parameter sets from a previous calibration and run model simulations.
-e	string	Path to the output directory

Table 2: Description of input and output files

File name	Dimension (rows x columns) ^a	Type	Sep	Description
INPUT				

parameters	11x2	Column 1: string Column 2: double/int Parameter details: OMEGA: double GAMMA: double NSIM: int BETASTART: double OVERDISPSTART: double REPORTINGSTART: double SEEDINFSTART: double SD_BETA: double SD_OVERDISP: double SD_REPORTING: double SD_SEEDINF: double	Tab	Column 1: parameter name Column 2: parameter value Parameter details: • OMEGA: rate of progression from exposed to infectious individuals (weeks⁻¹) • GAMMA: recovery rate (weeks⁻¹) • NSIM: Number of MCMC iterations • BETASTART: starting point for the scaling factor for transmissibility • OVERDISPSTART: starting point for the overdispersion parameter of the negative binomial likelihood • REPORTINGSTART: starting point for reporting ratio • SEEDINFSTART: starting point for number of infectious individuals at the start of the simulation • SD_BETA: standard deviation of the Gaussian proposal distribution for scaling factor for transmissibility • SD_OVERDISP: standard deviation of the Gaussian proposal distribution for the overdispersion parameter • SD_REPORTING: standard deviation of the Gaussian proposal distribution for the reporting ratio • SD_SEEDINF: standard deviation of the Gaussian proposal distribution for the number of infectious individuals at the start of the simulation
ILI_A_2023_2024	Wx3	double	Tab	Column 1: week of the 2023/2024 influenza season Column 2: weekly incidence of influenza-like illnesses per 1,000 related to influenza A viruses
matrix_IT_presence	AxA	double	Tab	Average number of contacts of an individual attending work/school in-person of age group a (row) with individuals of age group $\tilde{a}$ (column)
matrix_IT_not_presence	AxA	double	Tab	Average number of contacts of an individual not attending work/school in-person of age group a (row) with individuals of age group $\tilde{a}$ (column)
pop_by_age	Ax1	Int	Tab	Population size by age group a (row) in the Italian population
pop_by_age_presence	Ax1	Int	Tab	Number of individuals of age a (row) attending work/school in-person
pop_by_age_not_presence	Ax1	Int	Tab	Number of individuals of age a (row) not attending work/school in-person
susc_age_input	Ax1	double	Tab	Relative susceptibility to infection of age group a compared to the reference group ($a=1$)
OUTPUT				
CumInfByAgeP_* * runs from 0 to NSIM-1	AxW	double	Tab	Number of new infections in individuals attending school/work in-person (age group a , week w)

CumInfByAgeNP_* * runs from 0 to NSIM-1	AxW	double	Tab	Number of new infections in individuals not attending school/work in-person (age group <i>a</i> , week <i>w</i>)
mcmcprogress	NSIMx7	-	Tab	MCMC chain summary Column 1: ID of the MCMC iteration (from 0 to NSIM-1) Column 2: accepted value of the scaling factor for transmissibility. Column 3: accepted value of the reporting ratio. Column 4: accepted value of the number of infectious individuals at the start of the simulation. Column 5: accepted value of the overdispersion parameter of the negative binomial likelihood. Column 6: Log-likelihood of the last accepted simulation. Column 7: ID of the last accepted simulation.

^a: W: number of simulation weeks (24; from week 46 of 2023 to week 17 of 2024); A: number of age groups (15).